

Determinant of the Bordered Hessian (with 3×3 Example)

Setup. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be maximized (or minimized) subject to m equality constraints $g(x) = 0$, where $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$. The Lagrangian is

$$\mathcal{L}(x, \lambda) = f(x) - \lambda^\top g(x), \quad g(x) = \begin{bmatrix} g_1(x) \\ \vdots \\ g_m(x) \end{bmatrix}.$$

At a stationary point (x^*, λ^*) , define the constraint Jacobian and the Lagrangian Hessian

$$G = \nabla g(x^*) \in \mathbb{R}^{m \times n}, \quad H = \nabla_{xx}^2 \mathcal{L}(x^*, \lambda^*) \in \mathbb{R}^{n \times n}.$$

The *bordered Hessian* is the $(m+n) \times (m+n)$ block matrix

$$B = \begin{bmatrix} \mathbf{0}_{m \times m} & G \\ G^\top & H \end{bmatrix}.$$

Determinant via block/Schur complement. Assume H is invertible. Using the block determinant identity,

$$\det \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \det(D) \det(A - BD^{-1}C),$$

with $A = \mathbf{0}$, $B = G$, $C = G^\top$, $D = H$, we get

$$\det(B) = \det(H) \det(A - BD^{-1}C) = \det(H) \det(-GH^{-1}G^\top) = (-1)^m \det(H) \det(GH^{-1}G^\top).$$

For $m = 1$ (one constraint), writing $G = [g_1 \ \cdots \ g_n]$,

$$\det(B) = - \det(H) \ G H^{-1} G^\top.$$

Exact 3×3 bordered Hessian (one constraint, two variables). Take $n = 2$, $m = 1$. Write

$$H = \begin{bmatrix} a & b \\ b & c \end{bmatrix}, \quad G = [g_1 \ g_2].$$

Then the bordered Hessian is

$$B = \begin{bmatrix} 0 & g_1 & g_2 \\ g_1 & a & b \\ g_2 & b & c \end{bmatrix}.$$

The 2×2 Hessian determinant is $\Delta = \det(H) = ac - b^2$, and

$$H^{-1} = \frac{1}{\Delta} \begin{bmatrix} c & -b \\ -b & a \end{bmatrix}.$$

Hence

$$G H^{-1} G^\top = \frac{1}{\Delta} (c g_1^2 - 2b g_1 g_2 + a g_2^2), \quad \det(B) = -\det(H) \cdot G H^{-1} G^\top = -(c g_1^2 - 2b g_1 g_2 + a g_2^2).$$

Direct hand expansion check (same 3×3). Expanding $\det(B)$ along the first row:

$$\det(B) = 0 \cdot (\dots) - g_1 \cdot \det \begin{bmatrix} g_1 & b \\ g_2 & c \end{bmatrix} + g_2 \cdot \det \begin{bmatrix} g_1 & a \\ g_2 & b \end{bmatrix}.$$

Compute the 2×2 minors:

$$\det \begin{bmatrix} g_1 & b \\ g_2 & c \end{bmatrix} = g_1 c - b g_2, \quad \det \begin{bmatrix} g_1 & a \\ g_2 & b \end{bmatrix} = g_1 b - a g_2.$$

Therefore

$$\det(B) = -g_1(g_1 c - b g_2) + g_2(g_1 b - a g_2) = -c g_1^2 + 2b g_1 g_2 - a g_2^2 = -(c g_1^2 - 2b g_1 g_2 + a g_2^2),$$

which exactly matches the Schur complement result above.

Bordered principal minors and second-order tests (quick reference). For m equality constraints, define the bordered principal minors of order $m + k$ (border first, then successively add one decision variable):

$$\Delta_{m+k} = \det(\text{leading } (m+k) \times (m+k) \text{ principal submatrix of } B), \quad k = 1, \dots, n.$$

- **Maximization:** At a regular stationary point, a sufficient condition for a local maximum is

$$(-1)^m \Delta_{m+1} > 0, \quad (-1)^m \Delta_{m+2} > 0, \quad \dots, \quad (-1)^m \Delta_{m+n} > 0.$$

- **Minimization:** A sufficient condition for a local minimum is

$$(-1)^m \Delta_{m+1} < 0, \quad (-1)^m \Delta_{m+2} < 0, \quad \dots, \quad (-1)^m \Delta_{m+n} < 0.$$

These criteria use the signs of bordered minors rather than $\det(B)$ alone. When H is invertible, the Schur formula above is a fast way to compute $\det(B)$ (and the top-order minor) exactly.

Remarks.

- The ordering of rows/columns matters: the standard convention borders the constraints first, then appends the Hessian block H .
- If H is singular, the Schur complement formula is not applicable as written. In that case, compute $\det(B)$ directly (e.g., expansion or row/column operations that preserve determinants).
- For multiple constraints, $\det(GH^{-1}G^\top)$ is the squared volume (up to scaling) induced by the metric H^{-1} on the constraint gradients; the $(-1)^m$ factor tracks the m -fold sign from the zero block on the border.